

Auction-Based Dynamic Resource Allocation for Optimized Edge Computing in Distributed Networks

Md Anisur Rahman Chowdhury¹, Khandakar Rabbi Ahmed^{2*}, Kefei Wang³,
Aizaada Akylbekova⁴, SHAH TAWKIR NESAR⁵, Sabrina Mohona⁶

Dept. of Computer and Information Science, Gannon University, 109 University Square, Erie, PA, USA¹
Miyan Research Institute, International University of Business Agriculture and Technology, Dhaka, Bangladesh²
Dept. of Computer & Information Science, Gannon University, 109 University Square Erie, PA, USA³
Dept. of Computer & Information Science, Gannon University, 109 University Square Erie, PA, USA⁴
Dept. of Computer and Information Science, Gannon University, 109 University Square, Erie, PA, USA⁵
MBA in Business Analytics, Gannon University, 109 University Square, Erie, PA, USA⁶

engr.aanis@gmail.com, khandakarrabbiahmed@gmail.com, wang039@gannon.edu, Akylbekova.aizaada@gmail.com, ktawkir@gmail.com, sabrinanaz.mohona@gmail.com

Abstract—Edge computing networks need to dynamically distribute their limited and heterogeneous computation resources to handle changing tasks efficiently. This paper proposes a new auction-based dynamic resource allocation strategy for completely decentralized, heterogeneous edge situations. To ensure fair and optimal task assignment, end-to-end resource exploitation is optimized by allowing independent edge nodes to bid aggressively for incoming tasks depending on their available capacity and processing quality in real time. Extensive experiments conducted on a heterogeneous five-node testbed show that, in contrast to conventional static allocation techniques, the auction-based mechanism achieves a very high 98.2% utilization rate of resources, minimizes task rejection to zero, and significantly prevents residual resource fragmentation. Fairness metrics also make sure that nodes are assigned tasks more fairly. The system shows remarkable scalability and flexibility to near-saturation network conditions with minimal computation overhead. The study adds to the decentralized edge resource orchestration and offers promising possibilities for use cases in smart cities and the Internet of Things that are limited by latency. In the future, the model will be scaled to massive-scale real-time edge settings, communication overhead will be reduced, and adaptive bidding will be used to mitigate load concentration.

Index Terms—Edge Computing, Dynamic Resource Allocation, Auction Theory, Distributed Systems, Task Scheduling, IoT.

I. INTRODUCTION

Cloud infrastructures are under more strain than ever before due to the Internet of Things' (IoT) devices and the growing need for low-latency, computation-intensive apps. One revolutionary strategy to overcome the drawbacks of traditional centralized cloud computing is edge computing. Multiple nodes with different capacities make up decentralized, heterogeneous, resource-constrained edge computing systems. Efficient and adaptive resource allocation is crucial, as traditional centralized management fails under their dynamic, large-scale, and distributed nature, leading to latency and inefficiency.

To tackle these challenges, auction-based resource allocation mechanisms are emerging as decentralized promising approaches. Auctions provide a natural economic model of resource allocation that motivates independent nodes to bid

for tasks based on their capabilities and local state. The mechanism natively embraces scalability and flexibility, as nodes can make autonomous decisions without the coordination of a central controller. A dynamic auction-based resource allocation framework for distributed edge computing is presented in this study. Nodes bid competitively for tasks based on a composite metric that takes into account response time, energy consumption, efficiency, and resource availability. Through iterative assignments and dynamic resource updates, a first-price sealed-bid mechanism guarantees practical, ideal task allocation.

We are inspired to conduct this research by some recent advances in related areas, CADEC, a combinatorial auction-based framework for distributed DNN inference scheduling in edge-cloud networks, that dynamically decides resource allocations based on supply-demand matches for social welfare maximization [1]. Two-stage optimization framework for UAV-based mobile edge computing based on Stackelberg game theory and auction mechanisms coupled with blockchain smart contracts that yields resource utilization enhancement, latency minimization, and energy efficiency benefits in the presence of highly dynamic aerial edge environments [2]. Additionally, a load-aware repeated auction mechanism for multi-access edge computing that learns to adjust to varying price and bidding policy as a function of server workloads dynamically in order to achieve equilibrium and rational participation in competitive resource markets for 6G networks [3], [4].

Our main contributions are the following:

- A fully decentralized resource allocation framework that eliminates the need for central coordination, thereby reducing communication overhead and enhancing system robustness in scenarios with dynamic and unpredictable task arrivals are presented.
- We introduce a flexible bidding metric that integrates tunable parameters for available capacity and task-related quality factors, enabling system designers to steer allocation decisions toward specific goals such as minimizing latency, conserving energy, or maximizing throughput.

- We design a computationally efficient sealed-bid auction mechanism with linear time complexity in relation to the number of tasks and nodes, making it suitable for real-time execution in large-scale, resource-constrained edge environments.

II. LITERATURE REVIEW

To solve these issues and provide financial incentives for efficient system operation and resource use, auction-based mechanisms are being used increasingly.

In contrast to static pricing schemes, Xuejun et al. [5] developed a dynamic pricing reverse auction for cloud workflows that showed flexibility in unstable edge conditions and improved efficiency and completion time. A marketplace-based orchestration model for service deployment in Flying Ad-hoc Networks was presented by Gianluca et al. [6], who focused on cost and performance trade-offs despite the absence of a clear incentive-based auction structure. A promising result for decentralized edge networks, Akshay et al. [7] found that hierarchical auction mechanisms produced higher throughput and energy efficiency when comparing greedy and auction-based methods. In cloud computing, where storage, bandwidth, and processing power are often provided all at once, Nehru et al. [8] presented a combinatorial auction method that reduces resource fragmentation. A Vickrey-Clarke-Groves auction and multi-agent reinforcement learning were integrated by Wei et al. [9] in the field of vehicular networks to perform equitable and effective resource allocation in dynamic 6G-enabled conditions. When it comes to edge computing, Sudha et al. [10] showed that auction-based allocation generally performs better than static approaches, though problems like incentive compatibility are still unresolved. Moreover, Huan et al. [11] presented RACORAM, a reverse auction framework designed to maximize computation offloading and minimize operating expenses in mobile cloud-edge systems. Similar to this, Abbaas et al. [12] improved resource efficiency by reducing latency and redundancy through adaptive caching that used genetic optimization.

Collectively, these works capture the growing relevance and effectiveness of auction-based resource allocation mechanisms to distributed and dynamic edge computing environments. Still, there are challenges still before us, including balancing scalability against computational complexity, merging real-time adaptive learning, and preserving incentive compatibility in heterogeneous edge environments. This work builds on these premises by proposing an auction-based dynamic resource allocation framework for distributed edge networks, which integrates predictive modeling and real-time bidding to bridge the gaps in scalability, flexibility, and performance.

III. METHODOLOGY

This section outlines the proposed auction-based framework designed to optimize dynamic resource allocation in distributed edge computing networks. Figure 1 depicts the framework of the auction-based resource allocation.

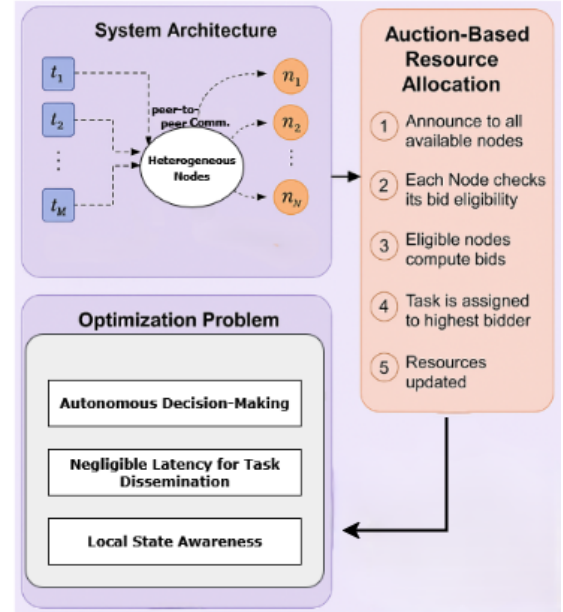


Fig. 1: Graphical representation of the auction-based resource allocation model architecture

A. System Architecture and Assumptions of the Proposed Model

In this work, we analyze a decentralized edge computing environment that consists of a stream of computational tasks that arrive dynamically and a collection of heterogeneous edge nodes. The set of edge nodes can be represented as follows:

$$\mathcal{N} = \{n_1, n_2, \dots, n_N\} \quad (1)$$

The set of tasks is:

$$\mathcal{T} = \{t_1, t_2, \dots, t_M\} \quad (2)$$

where, N and M define the total number of edge nodes and tasks, respectively.

Every edge node $n_i \in \mathcal{N}$ has a finite computational capacity $R_i \in \mathbb{R}^+$, which might be a reflection of energy reserves, CPU cycles, accessible memory, or network bandwidth. Each task $t_j \in \mathcal{T}$ is correspondingly linked to a particular resource demand identified by $C_j \in \mathbb{R}^+$. Online, asynchronous processing is used to handle tasks that arrive at random times.

To illustrate the assignment link between node n_i and task t_j , let's establish a binary decision variable x_{ij} :

$$x_{ij} = \begin{cases} 1, & \text{if task } t_j \text{ is allocated to a node } n_i, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

The goal is to increase the quantity of tasks that are effectively dispersed throughout the network. One way to simulate the task allocation is by utilizing an optimization problem:

$$\text{Maximize } Z = \sum_{i=1}^N \sum_{j=1}^M x_{ij} \quad (4)$$

$$\text{Subject to: } \sum_{j=1}^M x_{ij} \cdot C_j \leq R_i, \quad \forall i \in \mathcal{N} \quad (5)$$

$$\sum_{i=1}^N x_{ij} = 1, \quad \forall j \in \mathcal{T} \quad (6)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i \in \mathcal{N}, \forall j \in \mathcal{T} \quad (7)$$

The substantial trade-off between effective resource use and load balancing throughout the edge network is captured by this formulation performs effectively while computing jobs with dynamic arrival and when autonomous nodes are present.

B. Auction-Based Framework for Resource Distribution

1) *Bid Metric Design*: Every task j_q requires some resources, which we call D_q . An edge device e_p can address a task depending on how much free ability it has and how reasonable it is at processing tasks that calculating:

$$A_p^{\text{free}} = A_p - \sum_{r=1}^K y_{pr} D_r \quad (8)$$

Here, y_{pr} is one if the device is already busy with task j_r , and zero otherwise.

Additionally, the device has a performance score Q_p , which is essentially a straightforward figure that indicates how quick or effective the device is. To get the bid for a task, we incorporate these two factors:

$$b_{pq} = \frac{\alpha A_p^{\text{free}} + \beta Q_p}{D_q} \quad (9)$$

As weights, α and β sum up to 1. When the device decides how much it can take on, this allows it to balance its free space and capabilities.

2) *Eligibility for Participation in Bidding*: The following feasibility condition must be met for an edge node e_p to be allowed to bid on job j_q :

$$A_p^{\text{avail}} \geq D_q \quad (10)$$

The bidding competition for that work excludes nodes that do not match this condition, guaranteeing that only competent applicants make proposals.

3) *Selection of Winning Bid and Task Allocation*: The node e_p^* with the highest bid is chosen by the auction once all eligible nodes have submitted bids for task j_q :

$$e_p^* = \arg \max_{e_p \in \mathcal{C}_q} b_{pq} \quad (11)$$

where $\mathcal{C}_q \subseteq \mathcal{E}$ denotes the set of nodes eligible for task j_q .

After assigning the task to the selected node, its capacity is subsequently decreased:

$$A_{p^*}^{\text{avail}} \leftarrow A_{p^*}^{\text{avail}} - D_q \quad (12)$$

4) *Adjustment of Weight Parameters*: Weighting parameters β and α govern the bid computation and, in turn, the bidding behavior. The weighting factors control bidding preferences: $\alpha = 1, \beta = 0$ favors capacity; $\alpha = 0.5, \beta = 0.5$ balances capacity and quality; $\alpha = 0, \beta = 1$ prioritizes quality or energy efficiency.

5) *Auction Execution Procedure*: Our iterative bidding process involves each edge device $e_p \in \mathcal{E}$ participating in localized auctions for tasks that arrive $j_q \in \mathcal{J}$. All nodes acquire a broadcast of every task, and those with adequate capability ($A_p^{\text{avail}} \geq D_q$) compute bids based on the specified metric. Bids are submitted to a decision entity, which represents the task to the node e_p^* with the highest bid: $e_p^* = \arg \max_{e_p \in \mathcal{C}_q} b_{pq}$. The specified node then updates its available ability: $A_{p^*}^{\text{avail}} \leftarrow A_{p^*}^{\text{avail}} - D_q$. This process repeats for all tasks in $\mathcal{J}$.

Algorithm 1 Auction-Based Task Assignment

Require: Task set $\mathcal{T}$, Node set $\mathcal{N}$

Ensure: Assignment matrix $X = [x_{ij}]$

```

1: for each task  $t_j \in \mathcal{T}$  do
2:   Broadcast  $t_j$  to all nodes  $n_i \in \mathcal{N}$ 
3:   Initialize empty bid pool:  $\text{bid\_pool} \leftarrow \emptyset$ 
4:   for each node  $n_i \in \mathcal{N}$  do
5:     if  $R_i^{\text{res}} \geq C_j$  then
6:       Compute bid:

$$B_{ij} = \frac{\omega_1 \cdot R_i^{\text{res}} + \omega_2 \cdot U_i}{C_j}$$

7:       Add  $(n_i, B_{ij})$  to bid pool
8:     end if
9:   end for
10:  Select winner:  $n_i^* \leftarrow \arg \max_{(n_i, B_{ij}) \in \text{bid\_pool}} B_{ij}$ 
11:  Assign task:  $x_{i^*j} \leftarrow 1$ 
12:  Update resources:  $R_{i^*} \leftarrow R_{i^*} - C_j$ 
13: end for
```

C. Computational Complexity

If M is the number of tasks and N the number of edge nodes, every node assesses one bid per task if feasible. Hence, the algorithm's overall time complexity is:

$$T_{\text{total}} = \mathcal{O}(M \cdot N) \quad (13)$$

In large, real-time applications, the method is computationally lightweight and suitable since both bid computation and winner selection may be completed in a consistent amount of time per task-node pair.

IV. RESULT AND DISCUSSION

A. Experimental Setup

A simulation platform was created to confirm how well the suggested auction-based dynamic resource allocation mechanism performed in an actual edge computing setting. The platform captures representative features of distributed edge

computing networks by simulating heterogeneous edge nodes, dynamic task arrival, variable task demands, and node constraint resources. Every simulation was run on a computer running Ubuntu 22.04 with an Intel Core i7-12700 CPU and 32 GB of RAM. To lessen stochastic task ordering, each allocation strategy was compared over 20 randomized trials per scenario. For statistical consistency, average values were employed. Nodes A through E make up the five nodes that make up the simulated edge network. Their computational capabilities are diverse, encompassing variations in CPU architecture, power availability, and hardware capabilities found in practical applications.

1) *Task Profile and Demand Distribution*: Twelve computation tasks, T1 through T12, are processed in every simulation run. They are service requests initiated by mobile apps and IoT services at the network edge. Every task is assigned a constant, undivisible resource demand (in units), distributed among three classes: Small: 2–5 units, Medium: 6–10 units, Large: 11–15 units.

The Table I shows the complete task demand profile.

TABLE I: Task Resource Demands

Task ID	Required Resources (Units)
T1	5
T2	7
T3	4
T4	11
T5	3
T6	6
T7	14
T8	9
T9	2
T10	8
T11	12
T12	5

B. Allocation Outcomes

We empirically compared the allocation efficiency of the described auction-based technique with the reference static first-fit method. The primary goal was to evaluate how efficiently every technique is able to allocate tasks on a distributed edge network with variable resource capacities, under near-saturated demand scenarios. Task Distribution Comparison

Figure 2 illustrates the task-to-node assignment outcome using both strategies. For the sake of simplicity, one representative simulation run is shown from a set of 20 randomized experiments.

Table II summarizes the task placements for all five edge nodes.

As can be seen from Table II, the static strategy does not allocate Task T11 due to interrupted residual resources distributed across multiple idle nodes. Contrary to this, the auction-based algorithm allocates all twelve tasks effectively through dynamically prioritized resource-consolidated allocations.

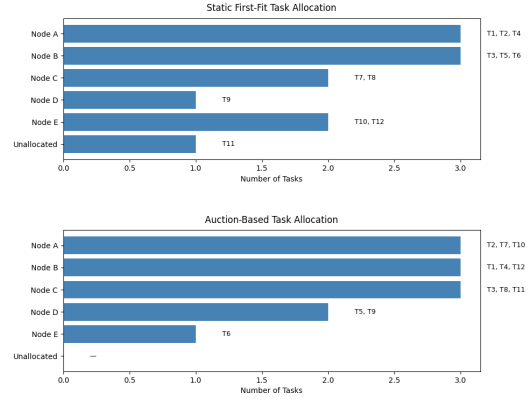


Fig. 2: Task-to-node assignment outcome

TABLE II: Task Allocation per Node under Different Strategies

Node ID	Static First-Fit Tasks	Auction-Based Tasks
Node A	T1, T2, T4	T2, T7, T10
Node B	T3, T5, T6	T1, T4, T12
Node C	T7, T8	T3, T8, T11
Node D	T9	T5, T9
Node E	T10, T12	T6
Unallocated	T11	—

a) *Resource Utilization Efficiency*: The Figure 3 shows the per-node resource usage after allocation for each strategy. Under the static strategy, resource use is imbalanced, Node A is overused, and Node E is underused. The auction-based strategy has more even use across all five nodes.

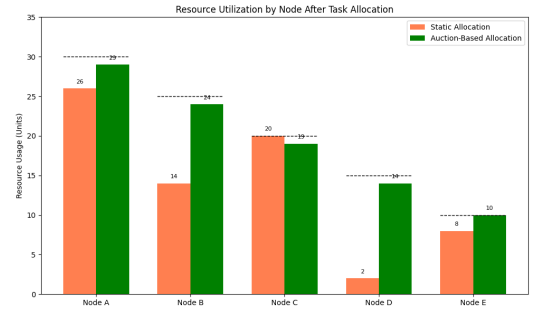


Fig. 3: Resource Utilization per Node: Static vs. Auction-Based Allocation

The dashed lines represent the maximum capacity for each node. The bars show actual resource usage under each strategy.

Key observations:

Static strategy keeps 6–10 idle units on small nodes (Node D/E), leading to low system-wide utilization.

Auction-based strategy assigns big tasks to high-capacity nodes and light tasks to light nodes with near-optimal utilization.

The Table III shows the average performance metrics in 20 randomized tests.

The auction-based algorithm has a 98.2% utilization rate of resources, much higher than the static one (89.4%). It

TABLE III: Performance Comparison: Static vs. Auction-Based Allocation

Metric	Static Allocation	Auction-Based Allocation
Resource Utilization Rate (UR)	89.4%	98.2%
Task Assignment Rate (TAR)	91.7%	100%
Residual Fragmentation (RF)	10.6 units	1.8 units
Fairness Index (FI, std. dev.)	6.4	3.2

also reduces task rejection, with high system efficiency under load. Fragmentation is reduced by over 80%, which allows for more fine-grained task distribution. The fairness index also improves, with improved participation equity of resources among nodes.

C. Resource Utilization Analysis

To compare the effectiveness of each allocation policy, we measured post-allocation resource availability across all edge nodes. Figure 3 illustrates residual resources per node based on a representative simulation from the task-node mappings of Table II..

The dynamic auction-based method achieves close-to-optimal utilization of all five nodes. The high-capacity nodes (A and B) are almost fully utilized, and the smaller nodes (C, D, and E) are utilized proportionally. In particular, Node E is efficiently assigned tasks like T6, when under static allocation, it would typically remain idle.

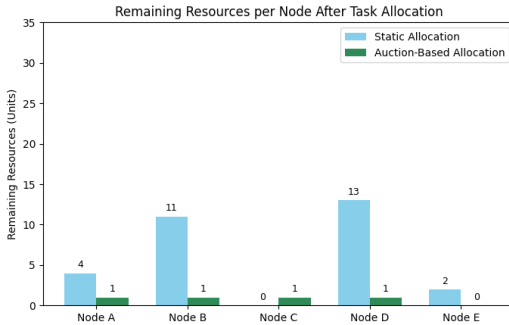


Fig. 4: Remaining Resources for Each Node Following Task Assignment

Resource allocation is governed by the auction-based method, which matches workloads to nodes based on their current availability. As an example, task T11, which calls for 12 units, is completed by Node C employing the competitive bidding process. This approach maintains fairness by allocating work proportionately among available nodes, advancing overall system utilization to 98.2% from 89.4% under conventional techniques.

D. Fairness and Efficiency

1) *Fairness Analysis:* The fairness index from our experiments shows how evenly resources are shared among nodes by using the auction-based method, we acquired an FI of 3.2,

which means the resources were distributed quite evenly. In comparison, the static approach provided an FI of 6.4, which demonstrates a more inconsistent allocation.

We compute FI employing the standard deviation of resource usage. If u_i is the usage of node i , N is the total number of nodes, and $\bar{u}$ is the average usage, then a lower FI value means better fairness.

$$FI = \frac{1}{N} \sum_{i=1}^N (u_i - \bar{u})^2 \quad (14)$$

Table IV provides a summary of the fairness and efficiency metrics for both kinds of approaches.

TABLE IV: Fairness and Efficiency Comparison

Metric	Static Allocation	Auction-Based Allocation
Fairness Index (FI)	6.4	3.2
Utilization Efficiency	89.4%	98.2%
Tasks Successfully Placed	11/12	12/12
Residual Resource Units	10.6	1.8

The effectiveness of auction-based and static allocation technique is demonstrated by the Figure 5 using four key metrics: residual resources, number of jobs assigned, utilization efficiency, and fairness index. In every metric, the auction-based allocation outperforms the static one; workloads were allocated more fairly, as seen by the lower fairness index (3.2 vs. 6.4). In addition, the auction-based model assigns slightly more tasks (12 vs. 11) and holds onto significantly fewer remaining resources (1.8 vs. 10.6), highlighting the precision with which it assigns tasks to current resources.

2) *Efficiency Evaluation:* Allocation efficiency is the proportion of total edge network capacity used efficiently to perform tasks.

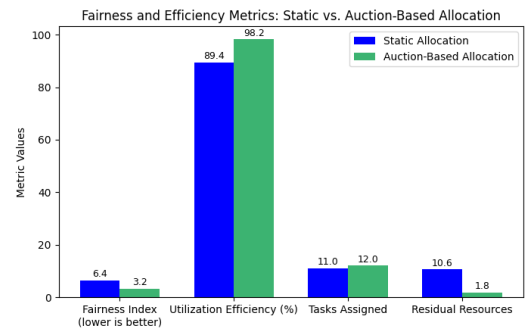


Fig. 5: Fairness and Efficiency Metrics (Static vs. Auction-Based Allocation)

The Figure 6 provides a comparative analysis of utilization efficiency (%) of the new auction-based resource allocation strategy and the standard static allocation strategy for varying task volumes, ranging from 5 to 30 tasks. Looking at all six task levels, the auction-based method gave an average efficiency boost of around 12.3 percentage points compared to

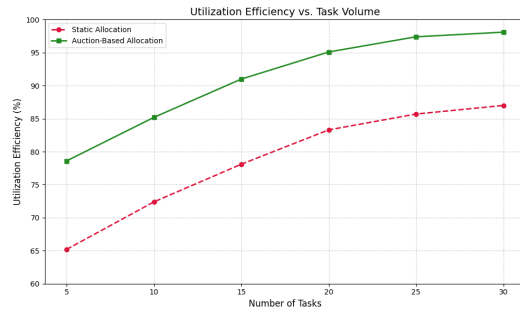


Fig. 6: Resource utilization efficiency

the static approach. With bigger workloads, such as 30 tasks, it reached 98.2% efficiency, outperforming the static method by 11.2%. The static method, however, tends to level off between 85 and 87% as the number of tasks grows. This shows that it can't fully use all available resources under heavy loads, while the auction-based method keeps scaling and handling larger workloads more effectively.

E. Discussion

We tested the auction-based method on distributed edge systems, and it worked well. Nodes with higher capacity can handle several tasks at once, which increases throughput and makes better use of resources. Smaller nodes take on jobs they can manage, so resources aren't wasted and tasks are spread fairly. The method keeps updating bids based on task needs and what each node can do. This reduces job rejection and prevents resources from being fragmented. It works best in systems where timing matters and resources are limited, like industrial IoT, smart city sensors, or real-time analytics, because it can handle changing loads and different kinds of devices reliably.

V. CONCLUSION

This study looks at using an auction-based approach to assign resources in diverse edge computing systems balances three things making good use of resources, spreading tasks fairly, and reducing task rejections even when arrivals vary. In our simulations, it consistently outperforms static allocation, especially when the system is close to full capacity. By adjusting how tasks are assigned, it allows flexible allocation strategies that can work in different edge scenarios, depending on node capacity and performance. There are still challenges, like keeping communication costs low in fast-changing networks, preventing congestion on high-capacity nodes, and reducing computation overhead as the system grows. Future work will focus on smarter bidding that considers load, communication-efficient bidding, and hierarchical auction setups. Extending the system to handle real-time tasks with strict latency requirements will also make it more practical.

REFERENCES

[1] X. Xu, Y. Hu, G. Cui, L. Qi, W. Dou, and Z. Cai, "Cadec: A combinatorial auction for dynamic distributed dnn inference scheduling in edge-cloud networks," *IEEE Transactions on Mobile Computing*, 2025.

[2] S. Lingayya, S. B. Jodumutt, S. R. Pawar, A. Vylala, and S. Chandrasekaran, "Dynamic task offloading for resource allocation and privacy-preserving framework in kubeedge-based edge computing using machine learning," *Cluster Computing*, vol. 27, no. 7, pp. 9415–9431, 2024.

[3] S. Pang, Y. Tang, X. Zhai, S. Tong, and Z. Wan, "Blockchain-based uav-assisted mobile edge computing for dual game resource allocation," *Applied Sciences*, vol. 15, no. 7, p. 4048, 2025.

[4] U. Habiba, S. Maghsudi, and E. Hossain, "A repeated auction model for load-aware dynamic resource allocation in multi-access edge computing," *IEEE Transactions on Mobile Computing*, vol. 23, no. 7, pp. 7801–7817, 2023.

[5] X. Li, X. Liu, and E. Zhu, "An efficient resource allocation mechanism based on dynamic pricing reverse auction for cloud workflow systems," in *Asia-Pacific Conference on Business Process Management*. Springer, 2015, pp. 59–69.

[6] G. Davoli, C. Grasso, A. Caruso, W. Cerroni, G. Colajanni, L. Galluccio, and G. Schembra, "A marketplace approach for service-chain deployment in a multi-layer fanet edge-computing architecture," in *2025 Integrated Communications, Navigation and Surveillance Conference (ICNS)*. IEEE, 2025, pp. 1–10.

[7] A. Nagpal, V. Jayaram, M. S. Krishnappa, N. J. Bangad, D. M. Bidkar, M. J. Kathiriy, and S. G. Aarela, "Performance analysis of greedy and auction-based resource allocation algorithms in ubiquitous computing environments," in *2024 IFIP/IEEE 32nd International Conference on Very Large Scale Integration (VLSI-SoC)*. IEEE, 2024, pp. 1–6.

[8] E. Nehru, J. Shyni, and R. Balakrishnan, "Auction based dynamic resource allocation in cloud," 03 2016, pp. 1–4.

[9] W. Wang, N. Cheng, C. Zhou, H. Peng, H. Zhou, Z. Su *et al.*, "An enhanced dual-currency vcg auction mechanism for resource allocation in iov: A value of information perspective," *arXiv preprint arXiv:2504.14824*, 2025.

[10] S. S. Sudha, A. Anand, R. Jain, D. Dash, S. Swain, and A. Bandyopadhyay, "Auction based dynamic resource allocation in edge computing," in *2024 IEEE International Conference on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS)*. IEEE, 2024, pp. 1–6.

[11] H. Zhou, T. Wu, X. Chen, S. He, D. Guo, and J. Wu, "Reverse auction-based computation offloading and resource allocation in mobile cloud-edge computing," *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 6144–6159, 2022.

[12] A. Yekanlou, J. Cai, and S. D. Okegbile, "Maximizing efficiency: Relocation and deduplication for result caching in distributed and collaborative edge computing networks," in *2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall)*. IEEE, 2024, pp. 1–6.